School of Informatics & Cybersecurity - Assessment Submission Coversheet

Student Information

Student Number:	B00168689
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Module Information

Module Title:	Database Fundamental
Module Code:	COMP-H1029

Assessment Information

Assessment Title:	CA

Declaration

☒ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☒ I have completed the TU Dublin AI Driving License theory test.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/lta/lta-resources/artificial-intelligence-driving-license/

☒ I declare that this assessment submission is my own work.

☒ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

Student signature:	Roisin Aylward
Date:	10/11/2025

Task One - Database Design (40% Total)

Clearly identify all relations to be included in the database.

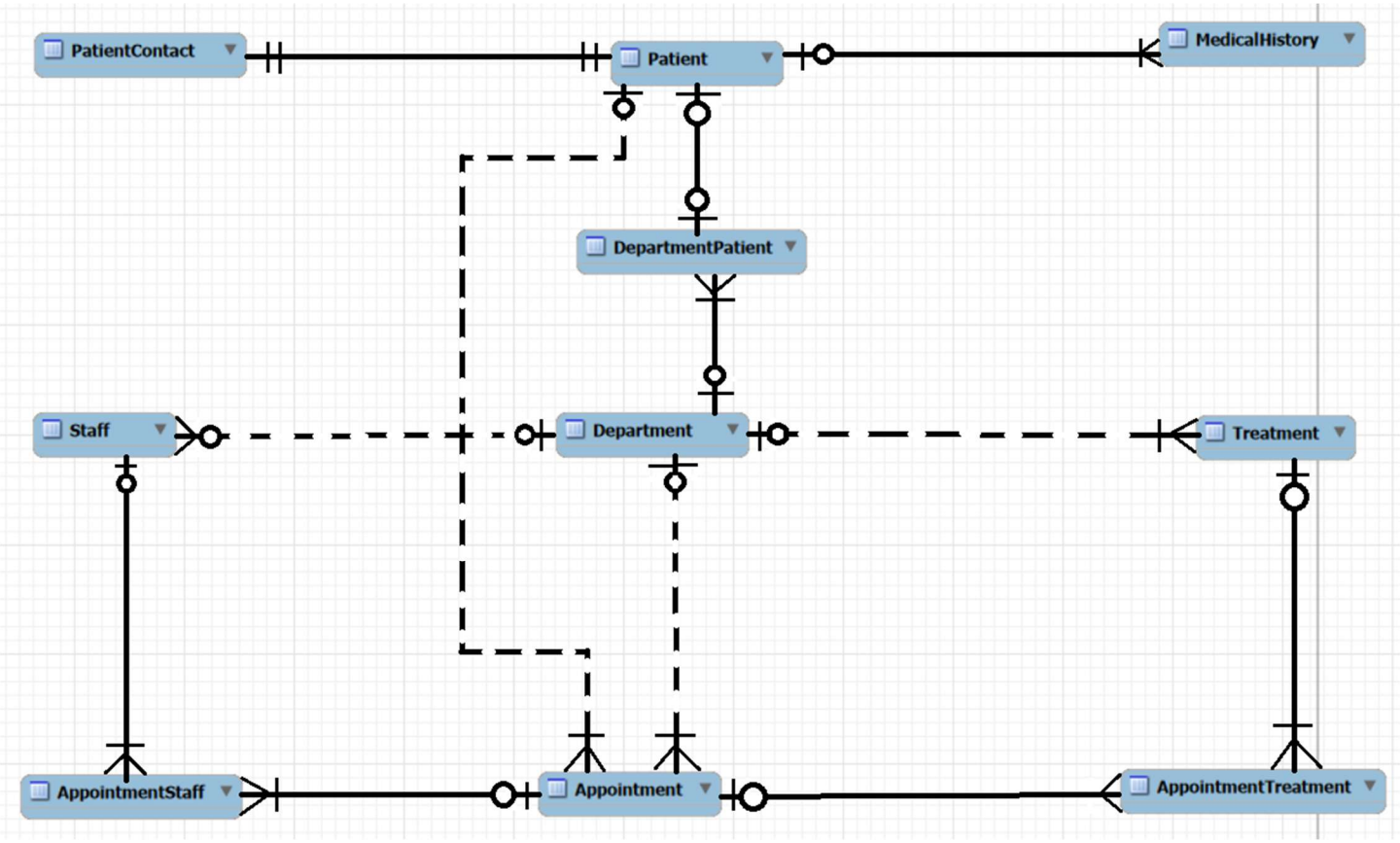
1. **Identify the relevant entities of the hospital dataset with their respective attributes, entity types (strong or weak) and primary keys. (8 Marks)**
   - Department (Strong):
     - DepartmentNo(PK)
     - DepartmentName
     - Location
     - BedCapacity
     - Phone
     - Email
   - Treatment (strong):
     - TreatmentNo(PK)
     - TreatmentName
     - DepartmentNo(FK from Department)
   - Staff (Strong entity type):
     - StaffID(PK)
     - StaffForeName
     - StaffSurName
     - Job
     - DepartmentNo(FK from Department)
     - Phone
     - Email
     - HireDate
     - DaysOff
   - Patient (strong entity type):
     - PatientID(PK)
     - PatientForeName
     - PatientSurName
     - DateOfBirth
     - Sex
   - Patient_Contact (weak entity type):
     - PatientID (PK+FK from Patient)
     - PhoneNumber
     - Email
     - Address1
     - Address2
     - County
     - Eircode
   - Medical_History (weak entity type)
     - PatientID (PK+FK from Patient)
     - Conditions (PK)

    - RecordDate (PK)
    - ConditionStatus
- Department_Patient (Weak Entity)
    - DepartmentNo(PK, FK from department)
    - BedNo(PK)
    - PatientID(FK)
    - AdmissionDate
- Appointment:
    - AppointmentID(PK)
    - DepartmentNo(FK from department)
    - PatientID(FK from patient)
    - Date
    - Time
    - Status
- Appointment_Staff:
    - AppointmentID(PK, FK from Appointment)
    - StaffID(PK, FK from Staff)
- Appointment_Treatment:
    - AppointmentID(PK, FK from Appointment)
    - TreatmentNo (PK, FK from Treatment)

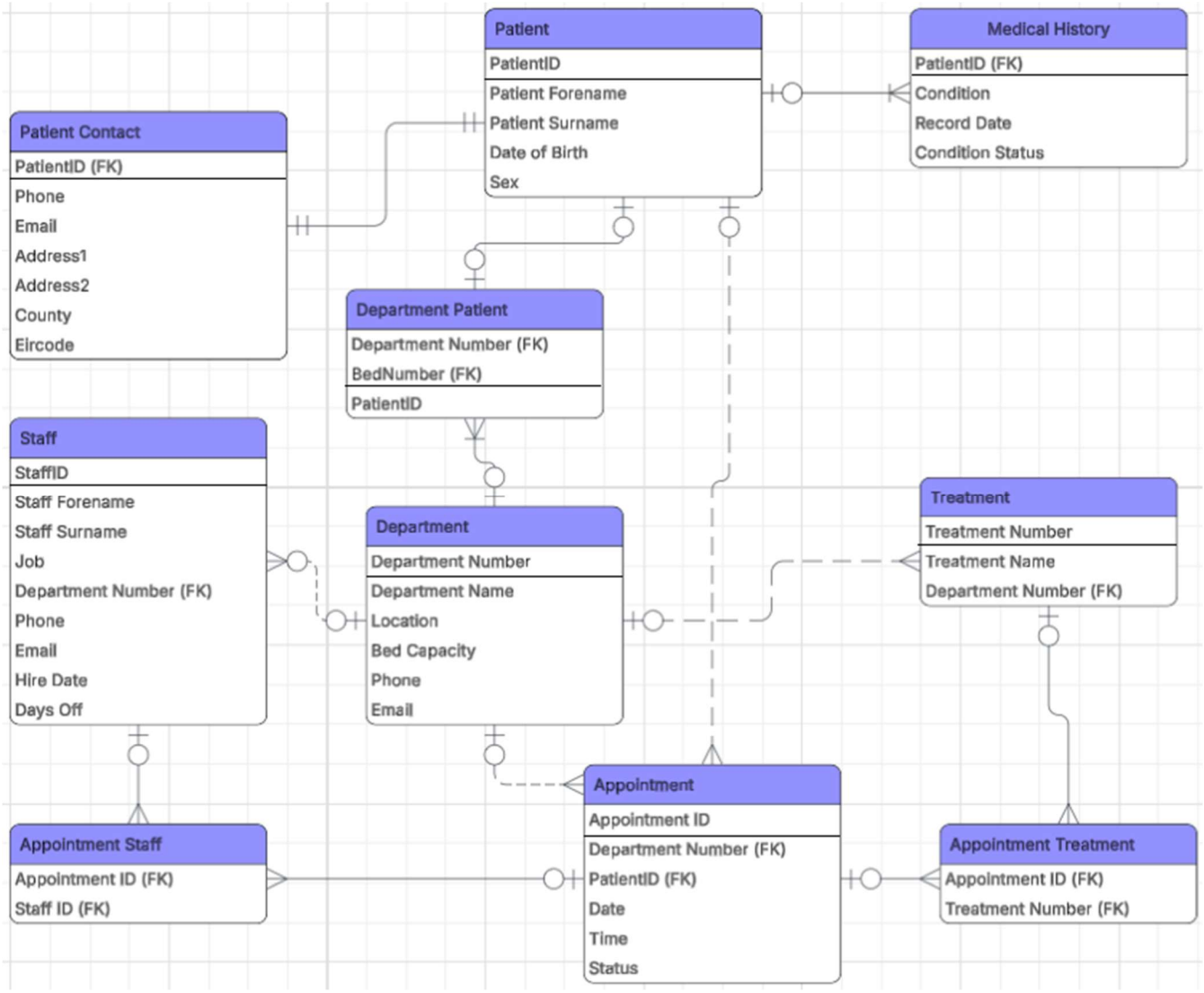
2. **Clearly explain why each relation (entity) needs to be included (8 Marks)**
    - Department lists the hospital medical departments, their contact information and their bed capacity as every medical department, except Administration, will have one ward attached to it.
    - Treatment lists all of the treatments that the hospital offers and the medical department that provides it.
    - Staff lists all of the staff working at the hospital, their jobs and the department they work for, when they were hired, and their days off. Enables appointment scheduling
    - Patient lists all of the patients and their basic information.
    - Patient_Contact lists the patients contact information. Required to have multiple address lines.
    - Medical_History lists each of the conditions that the patient have and when they got it. A patient can have more than one condition, therefore requiring its own table. A patient can also have two separate incidents of the same condition such as a fracture.
    - Dept_Patient lists all of the beds in every department and which patients are currently occupying them.
    - Appointment lists the patients appointment dates, times, and department.
    - Appointment_Staff lists the staff working in every appointment. An appointment can have more than one staff member.
    - Appointment_Treatment lists the treatment provided in an appointment. An appointment can have more than one treatment.
3. **Identify the relationships, cardinalities and participation constraints with supporting business rules and assumptions. (8 Marks)**

- Department – Treatment (1:N): One department can provide multiple treatments, but a treatment can only be provided by one department. It is optional for a department such as Administration to provide any treatment, but it is mandatory for a treatment to be provided by a department.
- Department – Staff (1:N): One department can have multiple staff working at it, but if a staff member is assigned to a department it can only be assigned to that one department. Participation is optional as a new department may initially have no staff, and certain staff members such as cleaners or cooks would not work for a particular department.
- Department – Appointment (1:N): One department can have multiple appointments, but a single appointment can only occur in one department. It is optional for a department such as Administration to have appointments, but it is mandatory for an appointment to happen in a department.
- Patient - Medical History (1:N): One patient can have multiple histories of multiple conditions, but every case of medical history can only be experienced by one patient. It is optional for a patient to have a medical history in cases where they are a new patient of the hospital awaiting test results, but it is mandatory for a case of a medical history to belong to a patient as its existence depends entirely on the patient.
- Patient - Patient_Contact (1:1): While a patient could have multiple phone numbers, emails, or addresses, only one of each is necessary and will be recorded by the hospital for simplicity. As phone numbers and email addresses are unique, every contact tuple will belong to one patient. Both relationships are mandatory as the Hospital needs to be able to contact the patients, and Patient_Contact's existence entirely depends on Patient's.
- Department - Dept_Patient (1:N): One department can have multiple beds in its ward, but a particular bed (Dept_Patient) can only be found in one department/ward. It is optional for a department to have beds in its ward in the case of the Administration Department or a new department that has yet to have beds, but it is mandatory for a bed (Dept_Patient) to be in a department.
- Patient - Dept_Patient (1:1): One patient can only be on a single bed at a given moment, and a bed can only have one patient currently occupying it. It is optional for a patient to be in a bed during intake, and it is optional for a bed to be occupied by a patient in cases of vacancy.
- Patient – Appointment (1:N): One patient can only attend a single appointment, but a patient can have multiple appointments. It is optional for a patient to have an appointment, but it is mandatory for an appointment to be scheduled for a patient.
- Appointment - Staff (M:N): An appointment can have multiple staff providing care, and a staff member can provide multiple care in multiple appointments. In a hospital with no waiting lists, it is optional for an appointment to be scheduled without immediately assigning staff to it, and it is optional for a staff member to have appointments as some such as cleaners would not.
- Appointment - Treatment (M:N): An appointment can have multiple treatments during it, and a treatment be provided in multiple appointments. In a hospital with no waiting lists, it is optional for an appointment to have treatment done during it as it could be solely an assessment/consultation, and it is optional for a treatment to be in any of the appointments.

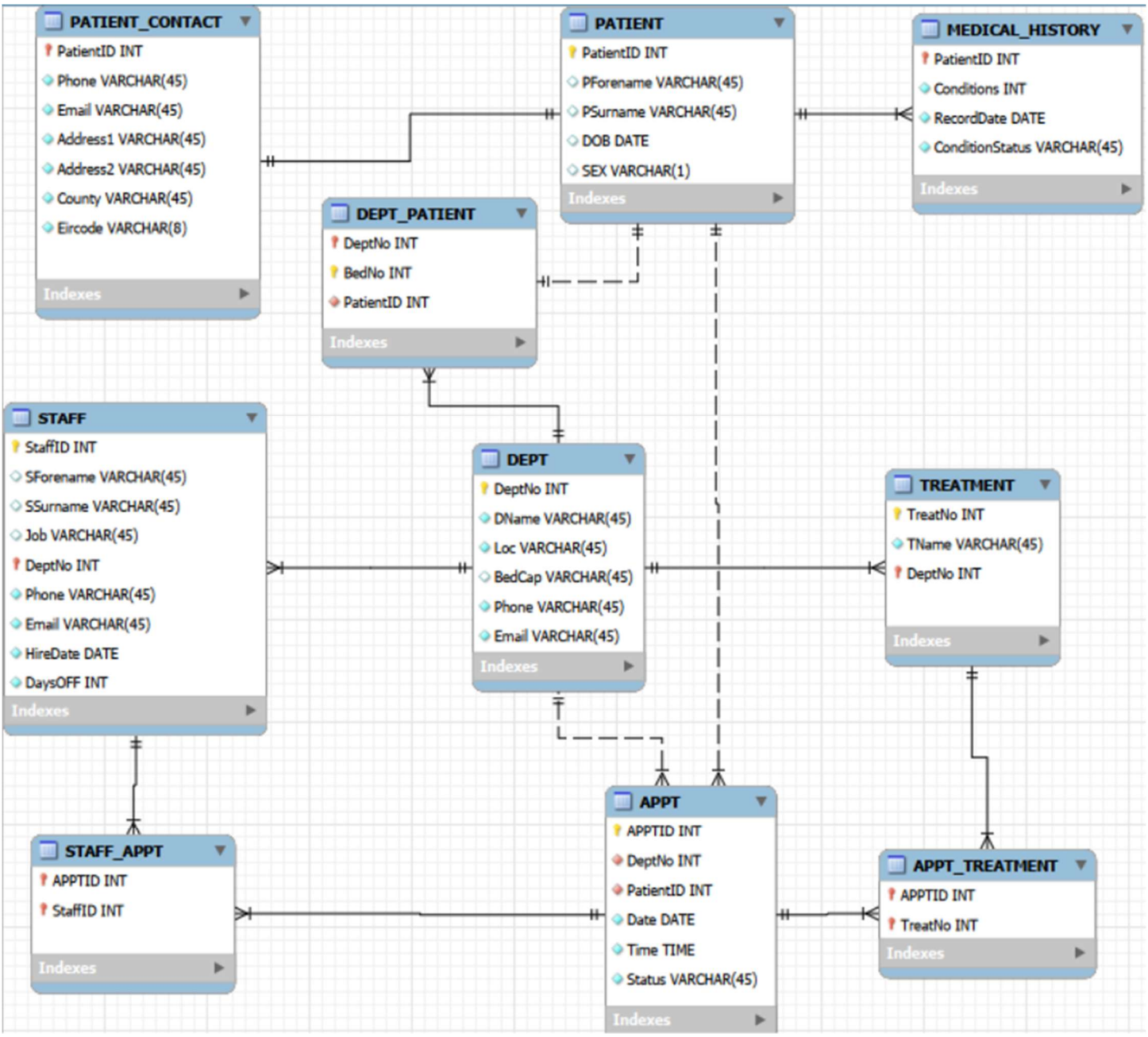
4. **Draw conceptual level ER diagram for the system depicting the entities, relationships, cardinalities, participation using your preferred ERD notation. (4 Marks)**



5. **Convert the entities of conceptual design into logical dataset and draw logical ERD. Use a logical layout of entities which is easily understood by any reader. (4 Marks)**



6. **Convert the entities of the logical design into physical model and draw physical level ERD. (4 Marks)**



7. **Convert the physical level ERD into relational model. Ensure that tables are in 3rd Normal Form. (4 Marks)**

Department(DeptNo(PK), TreatNo, DName, Loc, BedCap, Phone, Email)

Treatment(TreatNo(PK), TName, DeptNo(FK))

Staff(StaffID(PK), SForename, SSurname, Job, DeptNo(FK), Phone, Email, HireDate, DaysOff)

Patient(PatientID(PK), PForename, PSurname, DOB, Sex)

Patient_Contact(PatientID(PK, FK), Phone, Email, Address1, Address2, County, Eircode)

Medical_History(PatientID(PK, FK), ConditionName(PK), RecordDate(PK), Status)

Department_Patient(DeptNo(PK, FK), BedNo(PK), PatientID(FK), AdmissionDate)

Appointment(ApptID(PK), DeptNo(FK), PatientID(FK), Date, Time, Status)

Appointment_Staff(ApptID(PK, FK), StaffID(PK, FK))

Appointment_Treatment(ApptID(PK, FK), TreatNo(PK, FK))

Task Two – Create & Populate the database

Use DDL language to:

1. **Create the database and all tables specified in the dataset (relational model) to meet the specifications identified in task one. Clearly document all DDL statements used in this process.(10 Marks)**

```
CREATE SCHEMA `hospital`;

use HOSPITAL;

DROP TABLE IF EXISTS DEPT;

DROP TABLE IF EXISTS TREATMENT;

DROP TABLE IF EXISTS STAFF;

DROP TABLE IF EXISTS PATIENT;

DROP TABLE IF EXISTS DEPT_PATIENT;

DROP TABLE IF EXISTS MEDICAL_HISTORY;

DROP TABLE IF EXISTS PATIENT_CONTACT;

DROP TABLE IF EXISTS APPT;

DROP TABLE IF EXISTS APPT_TREATMENT;

DROP TABLE IF EXISTS APPT_STAFF;


CREATE TABLE DEPT (
 DEPTNO          INT NOT NULL,
 DNAME           VARCHAR(20),
 LOC                          VARCHAR(12),
 BEDCAP                              INT,
 PHONE                               VARCHAR(9),
 EMAIL                        VARCHAR(30),
 CONSTRAINT DEPT_PRIMARY_KEY PRIMARY KEY (DEPTNO));
```

```
CREATE TABLE TREATMENT (
 TREATNO        INT NOT NULL,
 TNAME          VARCHAR(40),
 DEPTNO                         INT NOT NULL,
 CONSTRAINT TREATMENT_DEPTNO_FK FOREIGN KEY (DEPTNO) REFERENCES DEPT (DEPTNO),
 CONSTRAINT TREATMENT_PRIMARY_KEY PRIMARY KEY (TREATNO));

CREATE TABLE STAFF (
 STAFFID        INT NOT NULL,
 SFORENAME      VARCHAR(20),
 SSURNAME               VARCHAR(20),
 JOB            VARCHAR(30),
 DEPTNO                         INT,
 PHONE                          INT,
 EMAIL                  VARCHAR(30),
 HIREDATE       DATE,
 DAYSOFF                INT,
 CONSTRAINT STAFF_DEPTNO_FK FOREIGN KEY (DEPTNO) REFERENCES DEPT (DEPTNO),
 CONSTRAINT STAFF_PRIMARY_KEY PRIMARY KEY (STAFFID));

CREATE TABLE PATIENT (
 PATIENTID      INT NOT NULL,
 PFORENAME      VARCHAR(20),
 PSURNAME               VARCHAR(20),
 DOB                    DATE,
 SEX            VARCHAR(1),
```

CONSTRAINT PATIENT_PRIMARY_KEY PRIMARY KEY (PATIENTID));

```
CREATE TABLE DEPT_PATIENT (
 DEPTNO                          INT NOT NULL,
 BEDNO                           INT NOT NULL,
 PATIENTID                INT,
 ADMISSION_DATE           DATE,
 CONSTRAINT DEPT_PATIENT_DEPT_FK FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO),
 CONSTRAINT DEPT_PATIENT_PATIENT_FK FOREIGN KEY (PATIENTID) REFERENCES PATIENT(PATIENTID),
 CONSTRAINT DEPT_PATIENT_PRIMARY_KEY PRIMARY KEY (DEPTNO,BEDNO));

CREATE TABLE MEDICAL_HISTORY (
 PATIENTID       INT NOT NULL,
 CONDITIONNAME           VARCHAR(40),
 RECORDDATE              DATE,
 CONDITIONSTATUS    VARCHAR(10),
 CONSTRAINT MEDICAL_HISTORY_PATIENT_FK FOREIGN KEY (PATIENTID) REFERENCES PATIENT (PATIENTID),
 CONSTRAINT MEDICAL_HISTORY_PRIMARY_KEY PRIMARY KEY (PATIENTID,CONDITIONNAME,RECORDDATE));

CREATE TABLE PATIENT_CONTACT (
 PATIENTID       INT NOT NULL,
 PHONE                           INT,
 EMAIL                    VARCHAR(35),
 ADDRESS1                 VARCHAR(30),
 ADDRESS2                 VARCHAR(30),
 COUNTY                          VARCHAR(9),
```

EIRCODE VARCHAR(8),

CONSTRAINT PATIENT_CONTACT_PATIENT_FK FOREIGN KEY (PATIENTID) REFERENCES PATIENT (PATIENTID),

CONSTRAINT PATIENT_PRIMARY_KEY PRIMARY KEY (PATIENTID));

CREATE TABLE APPT (

APPTID INT NOT NULL,

DEPTNO INT NOT NULL,

PATIENTID INT NOT NULL,

DATE DATE,

TIME TIME,

STATUS VARCHAR(20),

CONSTRAINT APPT_PRIMARY_KEY PRIMARY KEY (APPTID));

CREATE TABLE APPT_STAFF (

APPTID INT NOT NULL,

STAFFID INT NOT NULL,

CONSTRAINT APPT_STAFF_APPT_FK FOREIGN KEY (APPTID) REFERENCES APPT (APPTID),

CONSTRAINT APPT_STAFF_STAFF_FK FOREIGN KEY (STAFFID) REFERENCES STAFF (STAFFID),

CONSTRAINT APPT_PRIMARY_KEY PRIMARY KEY (APPTID,STAFFID));

CREATE TABLE APPT_TREATMENT (

APPTID INT NOT NULL,

TREATNO INT NOT NULL,

CONSTRAINT APPT_TREATMENT_APPT_FK FOREIGN KEY (APPTID) REFERENCES APPT (APPTID),

CONSTRAINT APPT_TREATMENT_TREATMENT_FK FOREIGN KEY (TREATNO) REFERENCES TREATMENT (TREATNO),

CONSTRAINT APPT_PRIMARY_KEY PRIMARY KEY (APPTID,TREATNO));

2. **Populate all database tables with the desired data to a minimum of 10 tuples per relation. Clearly document all DML statements used in this process (10 Marks)**

```
-- FORMAT: INSERT INTO DEPT VALUES (DEPTNO,'DNAME','LOC',BEDCAP,'PHONE','EMAIL');

INSERT INTO DEPT VALUES (1,'Cardiology','First Floor',4,'016466290','cardiology@hospital.ie');

INSERT INTO DEPT VALUES (2,'Dermatology','Second Floor',3,'018093958','dermatology@hospital.ie');

INSERT INTO DEPT VALUES (3,'Endocrinology','Third Floor',2,'014162661', 'endocrinology@hospital.ie');

INSERT INTO DEPT VALUES (4,'Gastroenterology','First Floor',2,'018032366','gastroenterology@hospital.ie');

INSERT INTO DEPT VALUES (5,'Urology','Third Floor',4,'012213125','urology@hospital.ie');

INSERT INTO DEPT VALUES (6,'Neurology','First Floor',3,'014284482','neurology@hospital.ie');

INSERT INTO DEPT VALUES (7,'Oncology','Second Floor',2,'016465639','oncology@hospital.ie');

INSERT INTO DEPT VALUES (8,'Orthopedics','First Floor',4,'017972214','orthopedics@hospital.ie');

INSERT INTO DEPT VALUES (9,'Otolaryngology','Second Floor',2,'018034859','otolaryngology@hospital.ie');

INSERT INTO DEPT VALUES (10,'Pediatrics','Fourth Floor',4,'015106100','pediatrics@hospital.ie');

INSERT INTO DEPT VALUES (11,'Administration','Third Floor',NULL,'012214000','admin@hospital.ie');


-- FORMAT: INSERT INTO TREATMENT VALUES (TREATNO,'TNAME',DEPTNO);

INSERT INTO TREATMENT VALUES (101,'Heart surgery',1);

INSERT INTO TREATMENT VALUES (102,'Blood transfusion',1);

INSERT INTO TREATMENT VALUES (201,'Skin grafting',2);

INSERT INTO TREATMENT VALUES (202,'Cryotherapy',2);

INSERT INTO TREATMENT VALUES (301,'Insulin',3);

INSERT INTO TREATMENT VALUES (401,'General surgery',4);

INSERT INTO TREATMENT VALUES (501,'Extracorporeal shockwave therapy',5);

INSERT INTO TREATMENT VALUES (601,'Neurostimulation',6);

INSERT INTO TREATMENT VALUES (701,'Chemotherapy',7);
```

INSERT INTO TREATMENT VALUES (702,'Radiotherapy',7);

INSERT INTO TREATMENT VALUES (801,'Joint replacement',8);

INSERT INTO TREATMENT VALUES (802,'Cartilage repair',8);

INSERT INTO TREATMENT VALUES (901,'Ear syringing',9);

INSERT INTO TREATMENT VALUES (902,'Tonsillectomy',9);

INSERT INTO TREATMENT VALUES (1010,'Vaccinations',10);

-- FORMAT: INSERT INTO STAFF VALUES
(STAFFID,'SFORNAME','SSURNAME','JOB',DEPTNO,PHONE,'EMAIL','HIREDATE',DAYSOFF);

INSERT INTO STAFF VALUES (11001,'Jason','Wade','CEO',11,013427843,'ceo@hospital.ie','2017-01-23',4);

INSERT INTO STAFF VALUES (1001,'Sarah','Brennan','Doctor',1,017213373,'sarah.brennan@hospital.ie','2009-07-08',2);

INSERT INTO STAFF VALUES (2001,'Adam','Dunne','Doctor',2,011273273,'adam.dunne@hospital.ie','2011-08-28',1);

INSERT INTO STAFF VALUES (3001,'Peter','Wang','Doctor',3,014383247,'peter.wang@hospital.ie','2011-01-09',2);

INSERT INTO STAFF VALUES (4001,'Anna','Connolly','Doctor',4,017336467,'anna.connolly@hospital.ie','2013-08-02',4);

INSERT INTO STAFF VALUES (5001,'Liam','Lennon','Doctor',5,018327832,'liam.lennon@hospital.ie','2011-12-07',6);

INSERT INTO STAFF VALUES (6001,'Alice','Nolan','Doctor',6,014372463,'alice.nolan@hospital.ie','2019-03-14',4);

INSERT INTO STAFF VALUES (7001,'Kavin','Smith','Doctor',7,013747373,'kavin.smith@hospital.ie','2002-02-02',1);

INSERT INTO STAFF VALUES
(8001,'Fionn','McCartney','Doctor',8,013474824,'fionn.mccartney@hospital.ie','2016-09-23',3);

INSERT INTO STAFF VALUES (9001,'James','White','Doctor',9,017438243,'james.white@hospital.ie','2010-05-25',9);

INSERT INTO STAFF VALUES (10001,'Josh','Murphy','Doctor',10,014378245,'josh.murphy@hospital.ie','2012-10-16',3);

INSERT INTO STAFF VALUES (11002,'Holly','Jones','Office worker',11,013724327,'holly.smith@hospital.ie','2023-03-30',8);

INSERT INTO STAFF VALUES (11003,'Nora','Dwyer','Office worker',11,017379744,'nora.dwyer@hospital.ie','2024-02-02',11);

INSERT INTO STAFF VALUES (1111,'Lisa','Costello','Cleaner',NULL,018349374,'cleaner@hospital.ie','2018-06-18',8);

INSERT INTO STAFF VALUES (1002,'Emily','Johnson','Nurse',1,016466295,'emily.johnson@hospital.ie','2022-03-05', 5);

INSERT INTO STAFF VALUES (2002,'Olivia','Williams','Nurse',2,018093961,'olivia.williams@hospital.ie','2023-02-12', 4);

INSERT INTO STAFF VALUES (3002,'Sophia','Brown','Nurse',3,014162664,'sophia.brown@hospital.ie','2021-08-23', 3);

INSERT INTO STAFF VALUES (4002,'Isabella','Maher','Nurse',4,018032372,'isabella.maher@hospital.ie','2023-01-17', 6);

INSERT INTO STAFF VALUES (5002,'Scarlett','Moore','Nurse',5,012213140,'scarlett.moore@hospital.ie','2021-04-09',7);

INSERT INTO STAFF VALUES (6002,'Ava','King','Nurse',6,016466301,'ava.king@hospital.ie','2022-06-14',5);

INSERT INTO STAFF VALUES (7004,'Zoe','Scott','Nurse',7,018093973,'zoe.scott@hospital.ie','2021-09-01',4);

INSERT INTO STAFF VALUES (8002,'Grace','Taylor','Nurse',8,018032386,'grace.taylor@hospital.ie','2022-11-22',6);

INSERT INTO STAFF VALUES (9002,'Jade','Hughes','Nurse',9,017972249,'jade.hughes@hospital.ie','2021-06-08',3);

INSERT INTO STAFF VALUES (10002,'Joseph','Clark','Nurse',10,012213155,'joseph.clark@hospital.ie','2021-11-02',7);

INSERT INTO STAFF VALUES (3003,'Benjamin','Taylor','Healthcare Assistant',3,014162700,'benjamin.taylor@hospital.ie','2021-06-14',3);

INSERT INTO STAFF VALUES (5003,'Oliver','Johnson','Healthcare Assistant',5,012213160,'oliver.johnson@hospital.ie','2021-11-17',7);

INSERT INTO STAFF VALUES (6003,'Ella','Davis','Healthcare Assistant',6,014284530,'ella.davis@hospital.ie','2023-03-05',4);

INSERT INTO STAFF VALUES (7005,'Lucy','Martin','Healthcare Assistant',7,016465680,'lucy.martin@hospital.ie','2022-07-18',5);

INSERT INTO STAFF VALUES (8003,'Clara','Moore','Healthcare Assistant',8,017972260,'clara.moore@hospital.ie','2021-05-11',3);

INSERT INTO STAFF VALUES (10003,'Megan','Anderson','Healthcare Assistant',10,015106145,'megan.anderson@hospital.ie','2023-02-13',6);

-- FORMAT: INSERT INTO PATIENT VALUES (PATIENTID,'PFORNAME','PSURNAME','DOB','SEX');

INSERT INTO PATIENT VALUES (5612,'John','Bob','1942-07-31','M');

INSERT INTO PATIENT VALUES (3456,'Saoirse','Murphy','1990-03-21','F');

INSERT INTO PATIENT VALUES (3412,'Eoghan','O Sullivan','1978-11-10','M');

INSERT INTO PATIENT VALUES (4131,'Aoife','Byrne','1963-02-17','F');

INSERT INTO PATIENT VALUES (1312,'Seamus','Kelly','1982-08-24','M');

INSERT INTO PATIENT VALUES (3213,'Niamh','Ryan','1968-05-30','F');

INSERT INTO PATIENT VALUES (2131,'Ciaran','O Reilly','1975-09-14','M');

INSERT INTO PATIENT VALUES (1434,'Orla','Doyle','1987-12-03','F');

INSERT INTO PATIENT VALUES (4314,'Padraig','McCarthy','1983-06-22','M');

INSERT INTO PATIENT VALUES (4321,'Clodagh','Fitzgerald','1965-01-06','F');

INSERT INTO PATIENT VALUES (1746,'Darragh','Lynch','1930-04-17','M');

INSERT INTO PATIENT VALUES (4325,'Caitriona','Murphy','1952-11-19','F');

INSERT INTO PATIENT VALUES (2431,'Ian','O Neill','1986-09-30','M');

INSERT INTO PATIENT VALUES (1523,'Roisin','Gallagher','1951-07-02','F');

INSERT INTO PATIENT VALUES (5123,'Brendan','Hughes','1989-10-05','M');

INSERT INTO PATIENT VALUES (5312,'Siobhan','Walsh','1951-01-14','F');

INSERT INTO PATIENT VALUES (5235,'Aidan','Smith','1974-12-08','M');

INSERT INTO PATIENT VALUES (4323,'Grainne','O Farrell','1994-03-23','F');

INSERT INTO PATIENT VALUES (1322,'Conor','Murray','2018-11-02','M');

INSERT INTO PATIENT VALUES (1432,'Maeve','Devlin','1956-04-27','F');

INSERT INTO PATIENT VALUES (4235,'Tadhg','Kennedy','1964-08-17','M');

INSERT INTO PATIENT VALUES (6576,'Bridget','Morris','1990-01-01','F');

INSERT INTO PATIENT VALUES (6543,'Ronan','O Malley','1927-10-14','M');

INSERT INTO PATIENT VALUES (4765,'Fiona','Murphy','1992-02-05','F');

INSERT INTO PATIENT VALUES (8457,'Patrick','Doherty','2019-05-09','M');

INSERT INTO PATIENT VALUES (4866,'Eimear','Cunningham','2014-07-19','F');

INSERT INTO PATIENT VALUES (4657,'Declan','Whelan','2013-11-13','M');

INSERT INTO PATIENT VALUES (4675,'Nuala','Sweeney','1988-02-11','F');

INSERT INTO PATIENT VALUES (4573,'Dermot','Graham','1961-12-22','M');

INSERT INTO PATIENT VALUES (4674,'Mairead','Horan','1991-09-03','F');

-- INSERT INTO DEPT_PATIENT VALUES (DEPTNO,BEDNO,PATIENTID,ADMISSION_DATE);

INSERT INTO DEPT_PATIENT VALUES (3,1,5612,'2025-10-31');

INSERT INTO DEPT_PATIENT VALUES (1,1,3456,'2025-11-21');

INSERT INTO DEPT_PATIENT VALUES (2,1,3412,'2025-11-11');

INSERT INTO DEPT_PATIENT VALUES (5,1,4131,'2025-10-07');

INSERT INTO DEPT_PATIENT VALUES (8,1,1312,'2025-10-24');

INSERT INTO DEPT_PATIENT VALUES (5,2,3213,'2025-11-13');

INSERT INTO DEPT_PATIENT VALUES (7,1,2131,'2025-11-14');

INSERT INTO DEPT_PATIENT VALUES (1,2,1434,'2025-11-23');

INSERT INTO DEPT_PATIENT VALUES (6,1,4314,'2025-11-23');

INSERT INTO DEPT_PATIENT VALUES (7,2,4321,'2025-11-06');

INSERT INTO DEPT_PATIENT VALUES (6,2,1746,'2025-11-19');

INSERT INTO DEPT_PATIENT VALUES (8,2,4325,'2025-11-19');

INSERT INTO DEPT_PATIENT VALUES (6,3,2431,'2025-10-30');

INSERT INTO DEPT_PATIENT VALUES (2,2,1523,'2025-07-02');

INSERT INTO DEPT_PATIENT VALUES (5,3,5123,'2025-10-31');

INSERT INTO DEPT_PATIENT VALUES (8,3,5312,'2025-10-14');

INSERT INTO DEPT_PATIENT VALUES (3,2,5235,'2025-10-18');

INSERT INTO DEPT_PATIENT VALUES (4,1,4323,'2025-11-23');

INSERT INTO DEPT_PATIENT VALUES (10,1,1322,'2025-11-12');

INSERT INTO DEPT_PATIENT VALUES (8,4,1432,'2025-09-27');

INSERT INTO DEPT_PATIENT VALUES (1,3,4235,'2025-08-17');

INSERT INTO DEPT_PATIENT VALUES (9,1,6576,'2025-11-21');

INSERT INTO DEPT_PATIENT VALUES (2,3,6543,'2025-11-12');

INSERT INTO DEPT_PATIENT VALUES (5,4,4765,'2025-11-25');

INSERT INTO DEPT_PATIENT VALUES (10,2,8457,'2025-11-19');

INSERT INTO DEPT_PATIENT VALUES (10,3,4866,'2025-10-29');

INSERT INTO DEPT_PATIENT VALUES (10,4,4657,'2025-11-23');

INSERT INTO DEPT_PATIENT VALUES (1,4,4675,'2025-11-11');

INSERT INTO DEPT_PATIENT VALUES (4,2,4573,'2025-11-22');

INSERT INTO DEPT_PATIENT VALUES (9,2,4674,'2025-11-23');

-- FORMAT: INSERT INTO MEDICAL_HISTORY VALUES (PATIENTID,'CONDITIONNAME','RECORDDATE','STATUS');

INSERT INTO MEDICAL_HISTORY VALUES (5612,'Melanoma','2019-05-01','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (5612,'Diabetes','1992-08-21','Active');

INSERT INTO MEDICAL_HISTORY VALUES (3456,'Hypertension','2015-03-12','Active');

INSERT INTO MEDICAL_HISTORY VALUES (3412,'Erythroderma','2011-06-23','Active');

INSERT INTO MEDICAL_HISTORY VALUES (3412,'Osteoarthritis','2018-07-30','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (4131,'Chronic Kidney Disease','2020-01-18','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1312,'Gout','2015-05-22','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1312,'Diabetes','2000-10-10','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (3213,'Urinary Tract Infection','2012-11-01','Active');

INSERT INTO MEDICAL_HISTORY VALUES (2131,'Prostate Cancer','2019-09-15','Relapse');

INSERT INTO MEDICAL_HISTORY VALUES (1434,'Anemia','2018-07-10','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4314,'Epilepsy','2015-01-11','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4314,'High Cholesterol','2017-03-29','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4321,'Cervical Cancer','2018-06-17','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1746,'Parkinsons Disease','2010-11-25','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1746,'Hypertension','2015-05-06','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (1746,'Osteoporosis','2016-02-20','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (4325,'Chronic Back Pain','2017-09-13','Active');

INSERT INTO MEDICAL_HISTORY VALUES (2431,'Migraines','2021-04-20','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1523,'Psoriasis','2014-08-16','Active');

INSERT INTO MEDICAL_HISTORY VALUES (5123,'Urinary Tract Infection','2019-05-12','Active');

INSERT INTO MEDICAL_HISTORY VALUES (5123,'Hypothyroidism','2015-10-05','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (5312,'Rheumatoid Arthritis','2016-09-30','Active');

INSERT INTO MEDICAL_HISTORY VALUES (5235,'Chronic Fatigue Syndrome','2019-07-20','Active');

INSERT INTO MEDICAL_HISTORY VALUES (5235,'Diabetes','2005-11-15','Active');

INSERT INTO MEDICAL_HISTORY VALUES (5235,'Obesity','2020-02-22','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4323,'Acid Reflux','2020-01-10','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1322,'Arrhythmia','2019-05-09','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1322,'Cerebral Palsy','2019-11-25','Active');

INSERT INTO MEDICAL_HISTORY VALUES (1432,'Osteoarthritis','2014-04-14','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4235,'Gastroesophageal Reflux Disease','2019-06-10','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4235,'Hypertension','2017-08-21','Active');

INSERT INTO MEDICAL_HISTORY VALUES (6576,'Tonsillitis','2021-01-05','Active');

INSERT INTO MEDICAL_HISTORY VALUES (6543,'Erythroderma','2010-05-18','Active');

INSERT INTO MEDICAL_HISTORY VALUES (6543,'Sleep Apnea','2016-08-30','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (6543,'Psoriasis','2014-03-22','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (4765,'Chronic Kidney Disease','2017-09-20','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4765,'Diabetes','2015-04-08','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4765,'Hypertension','2019-11-17','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4765,'Heart Disease','2016-06-12','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (8457,'Scoliosis','2025-03-14','Active');

INSERT INTO MEDICAL_HISTORY VALUES (8457,'Diabetes','2024-08-05','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (8457,'Leukemia','2021-09-01','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (4866,'Urinary Tract Infection','2025-04-17','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4657,'Irritable Bowel Syndrome','2020-03-05','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4657,'Chronic Prostatitis','2018-07-24','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4675,'Melanoma','2018-05-30','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4675,'Osteoarthritis','2014-07-22','Stable');

INSERT INTO MEDICAL_HISTORY VALUES (4675,'Chronic Back Pain','2016-02-09','Remission');

INSERT INTO MEDICAL_HISTORY VALUES (4675,'Heart Failure','2020-09-13','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4573,'Crohns Disease','2025-11-12','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4674,'Vertigo','2018-10-14','Active');

INSERT INTO MEDICAL_HISTORY VALUES (4674,'Migraine','2017-11-07','Stable');

-- FORMAT: INSERT INTO PATIENT_CONTACT VALUES (PATIENTID,PHONE,'EMAIL','ADDRESS1','ADDRESS2','COUNTY','EIRCODE');

INSERT INTO PATIENT_CONTACT VALUES (5612,0891112233,'johnbob42@gmail.com','5 High Street','Coolmine Estate','Dublin','D15 F865');

INSERT INTO PATIENT_CONTACT VALUES (3456,0862345678,'saoirse.murphy88@yahoo.ie','2 Oak Avenue','Shandon Park','Cork','T12 C4D5');

INSERT INTO PATIENT_CONTACT VALUES (3412,0873456789,'eoghan.osullivan.mel@protonmail.com','13 Birch Road','Riverstown','Limerick','V94 G5H6');

INSERT INTO PATIENT_CONTACT VALUES (4131,0864567890,'aoifebyrne2013@gmail.com','21 Maple Lane','Knocknacarra','Galway','H91 J7K8');

INSERT INTO PATIENT_CONTACT VALUES (1312,0895678901,'seamus.kelly98@yahoo.com','17 Pine Court','Willowbrook','Waterford','X91 L3M4');

INSERT INTO PATIENT_CONTACT VALUES (3213,0866789012,'niamhryan1975@outlook.com','6 Cedar Drive','Kilmacud','Kilkenny','R95 N8O2');

INSERT INTO PATIENT_CONTACT VALUES (2131,0877890123,'ciaran.oreilly42@icloud.com','8 Elm Place','Kilmainham','Dublin','D06 X8R2');

INSERT INTO PATIENT_CONTACT VALUES (1434,0868901234,'orladoyle@outlook.com','4 Willow Mews','Abbeyfield','Derry','BT47 6X7');

INSERT INTO PATIENT_CONTACT VALUES (4314,0879012345,'padraigmccarthy123@yahoo.ie','25 Ash Avenue','The Quays','Donegal','F92 V3W8');

INSERT INTO PATIENT_CONTACT VALUES (4321,0860123456,'clodagh.fitzgerald@protonmail.com','30 Oakwood Crescent','Ballaghaderreen','Kerry','V92 X1Z2');

INSERT INTO PATIENT_CONTACT VALUES (1746,0871239876,'darragh_lynch1990@icloud.com','12 Chestnut Road','Ashford Village','Cavan','A82 B3C9');

INSERT INTO PATIENT_CONTACT VALUES (4325,0862349876,'caitrionamurphy69@yahoo.ie','16 Heather Way','Clonmel Estates','Clare','V95 H6K1');

INSERT INTO PATIENT_CONTACT VALUES (2431,0833459876,'ian.oneill88@outlook.com','19 Rose Gardens','Kilcarra','Meath','A83 W4X5');

INSERT INTO PATIENT_CONTACT VALUES (1523,0864569876,'roisin.gallagher76@yahoo.com','22 Oakwood Drive','Dundalk Park','Monaghan','H18 4N6');

INSERT INTO PATIENT_CONTACT VALUES (5123,0875679876,'brendan_hughes12@gmail.com','7 Holly Lane','Rockfield','Louth','A91 L9F2');

INSERT INTO PATIENT_CONTACT VALUES (5312,0866789876,'siobhan.walsh42@protonmail.com','11 Fern Road','Greenwich','Tipperary','E34 X5H3');

INSERT INTO PATIENT_CONTACT VALUES (5235,0877899876,'aidansmith33@yahoo.ie','18 Pine Hill','Bellinstown','Carlow','R93 F8T4');

INSERT INTO PATIENT_CONTACT VALUES (4323,0868909876,'grainneofarrell1994@icloud.com','14 Birchwood Crescent','Tullamore Heights','Laois','R32 H7N9');

INSERT INTO PATIENT_CONTACT VALUES (1322,0839019876,'conormurray121@outlook.com','9 Cedar Drive','Beechwood','Kilkenny','R95 E2B6');

INSERT INTO PATIENT_CONTACT VALUES (1432,0860129876,'maevedevlin54@gmail.com','20 Maple Lane','Ballinasloe','Roscommon','F42 P3D2');

INSERT INTO PATIENT_CONTACT VALUES (4235,0871236547,'tadhg.kennedy54@yahoo.com','15 Willow Grove','Ballybricken','Tipperary','E41 F2D3');

INSERT INTO PATIENT_CONTACT VALUES (6576,0862346547,'bridgetmorris7@yahoo.ie','22 Elm Street','Hillview','Louth','A92 G4F5');

INSERT INTO PATIENT_CONTACT VALUES (6543,0873456547,'ronanomalley89@protonmail.com','23 Oak Park','Rathmines','Clare','V95 B7T9');

INSERT INTO PATIENT_CONTACT VALUES (4765,0864566547,'fiona.murphy88@icloud.com','4 Pine Road','Portlaoise Estate','Dublin','D06 X8R2');

INSERT INTO PATIENT_CONTACT VALUES (8457,0895676547,'patrick.doherty@email.com','10 Chestnut Avenue','Lifford Gardens','Waterford','X91 N5T1');

INSERT INTO PATIENT_CONTACT VALUES (4866,0866786547,'eimear.cunningham11@yahoo.ie','8 Birch Lane','Kilbarrack','Kilkenny','R95 B8M7');

INSERT INTO PATIENT_CONTACT VALUES (4657,0877896547,'declan.whelan@outlook.com','5 Maple Street','Forest Hills','Limerick','V94 M4P3');

INSERT INTO PATIENT_CONTACT VALUES (4675,0868906547,'nuala.sweeney54@protonmail.com','12 Cedar Crescent','Carrig House','Cavan','A82 N6R2');

INSERT INTO PATIENT_CONTACT VALUES (4573,0899016547,'dermot.graham@email.com','30 Ash Road','Fernhill','Galway','H91 K1P8');

INSERT INTO PATIENT_CONTACT VALUES (4674,0890126547,'maireadhoran@outlook.com','14 Willow Park','Shandon Heights','Derry','BT48 9X7');

-- FORMAT: INSERT INTO APPT VALUES (APPTID,DEPTNO,PATIENTID,'DATE','TIME','STATUS');

INSERT INTO APPT VALUES (1,2,5612,'2025-11-14','11:30:00','Completed');

INSERT INTO APPT VALUES (2,3,5612,'2025-11-13','15:30:00','Completed');

INSERT INTO APPT VALUES (3,1,3456,'2025-11-25','12:30:00','Completed');

INSERT INTO APPT VALUES (4,2,3412,'2025-11-11','16:00:00','Completed');

INSERT INTO APPT VALUES (5,8,3412,'2025-10-07','09:30:00','Completed');

INSERT INTO APPT VALUES (6,8,1312,'2025-10-24','11:00:00','Completed');

INSERT INTO APPT VALUES (7,5,3213,'2025-11-13','10:30:00','Completed');

INSERT INTO APPT VALUES (8,7,2131,'2025-11-14','14:30:00','Completed');

```
INSERT INTO APPT VALUES (9,1,1434,'2025-11-23','13:00:00','Completed');
INSERT INTO APPT VALUES (10,6,4314,'2025-11-23','09:00:00','Completed');
INSERT INTO APPT VALUES (11,7,4321,'2025-11-06','12:00:00','Completed');
INSERT INTO APPT VALUES (12,6,1746,'2025-11-19','10:00:00','Completed');
INSERT INTO APPT VALUES (13,8,4325,'2025-11-19','09:30:00','Completed');
INSERT INTO APPT VALUES (14,6,2431,'2025-10-30','13:30:00','Completed');
INSERT INTO APPT VALUES (15,2,1523,'2025-07-02','11:30:00','Completed');
INSERT INTO APPT VALUES (16,5,5123,'2025-10-31','15:00:00','Completed');
INSERT INTO APPT VALUES (17,8,5312,'2025-10-14','14:00:00','Completed');
INSERT INTO APPT VALUES (18,3,5235,'2025-10-18','09:00:00','Completed');
INSERT INTO APPT VALUES (19,4,4323,'2025-11-23','10:30:00','Completed');
INSERT INTO APPT VALUES (20,10,1322,'2025-11-12','11:00:00','Completed');
INSERT INTO APPT VALUES (21,8,1432,'2025-09-27','09:30:00','Completed');
INSERT INTO APPT VALUES (22,1,4235,'2025-08-17','10:00:00','Completed');
INSERT INTO APPT VALUES (23,9,6576,'2025-11-21','12:30:00','Completed');
INSERT INTO APPT VALUES (24,2,6543,'2025-11-12','13:30:00','Completed');
INSERT INTO APPT VALUES (25,5,4765,'2025-11-25','15:30:00','Completed');
INSERT INTO APPT VALUES (26,10,8457,'2025-11-19','14:30:00','Completed');
INSERT INTO APPT VALUES (27,10,4866,'2025-10-29','09:00:00','Completed');
INSERT INTO APPT VALUES (28,10,4657,'2025-11-23','09:00:00','Completed');
INSERT INTO APPT VALUES (29,2,4675,'2025-11-11','10:30:00','No show');
INSERT INTO APPT VALUES (30,4,4573,'2025-11-22','11:00:00','No show');
INSERT INTO APPT VALUES (31,2,5612,'2025-11-25','09:00:00','No show');
INSERT INTO APPT VALUES (32,3,5612,'2025-11-14','10:30:00','No show');
INSERT INTO APPT VALUES (52,9,6576,'2025-11-22','10:30:00','Cancelled');
INSERT INTO APPT VALUES (53,2,6543,'2025-11-15','11:00:00','Cancelled');
```

INSERT INTO APPT VALUES (54,5,4765,'2026-01-15','09:00:00','Scheduled');

INSERT INTO APPT VALUES (55,10,8457,'2026-01-08','09:30:00','Scheduled');

INSERT INTO APPT VALUES (56,10,4866,'2025-12-02','10:00:00','Scheduled');

INSERT INTO APPT VALUES (57,10,4657,'2025-12-05','11:30:00','Scheduled');

INSERT INTO APPT VALUES (58,1,4675,'2025-11-27','13:00:00','Scheduled');

INSERT INTO APPT VALUES (59,4,4573,'2026-01-10','14:30:00','Scheduled');

-- FORMAT: INSERT INTO APPT_STAFF VALUES (APPTID,STAFFID);

INSERT INTO APPT_STAFF VALUES (1,2001);

INSERT INTO APPT_STAFF VALUES (1,2002);

INSERT INTO APPT_STAFF VALUES (2,3001);

INSERT INTO APPT_STAFF VALUES (2,3002);

INSERT INTO APPT_STAFF VALUES (3,1002);

INSERT INTO APPT_STAFF VALUES (4,2001);

INSERT INTO APPT_STAFF VALUES (5,8002);

INSERT INTO APPT_STAFF VALUES (5,8003);

INSERT INTO APPT_STAFF VALUES (6,8001);

INSERT INTO APPT_STAFF VALUES (7,5003);

INSERT INTO APPT_STAFF VALUES (8,7001);

INSERT INTO APPT_STAFF VALUES (8,7004);

INSERT INTO APPT_STAFF VALUES (8,7005);

INSERT INTO APPT_STAFF VALUES (9,1001);

INSERT INTO APPT_STAFF VALUES (10,6002);

INSERT INTO APPT_STAFF VALUES (11,7001);

INSERT INTO APPT_STAFF VALUES (11,7004);

INSERT INTO APPT_STAFF VALUES (12,6001);

```
INSERT INTO APPT_STAFF VALUES (13,8002);
INSERT INTO APPT_STAFF VALUES (13,8003);
INSERT INTO APPT_STAFF VALUES (14,6001);
INSERT INTO APPT_STAFF VALUES (14,6003);
INSERT INTO APPT_STAFF VALUES (15,2001);
INSERT INTO APPT_STAFF VALUES (15,2002);
INSERT INTO APPT_STAFF VALUES (16,5003);
INSERT INTO APPT_STAFF VALUES (17,8002);
INSERT INTO APPT_STAFF VALUES (18,3003);
INSERT INTO APPT_STAFF VALUES (19,4002);
INSERT INTO APPT_STAFF VALUES (20,10002);
INSERT INTO APPT_STAFF VALUES (21,8002);
INSERT INTO APPT_STAFF VALUES (22,1001);
INSERT INTO APPT_STAFF VALUES (23,9002);
INSERT INTO APPT_STAFF VALUES (24,2002);
INSERT INTO APPT_STAFF VALUES (25,5002);
INSERT INTO APPT_STAFF VALUES (26,10002);
INSERT INTO APPT_STAFF VALUES (27,10002);
INSERT INTO APPT_STAFF VALUES (28,10002);
INSERT INTO APPT_STAFF VALUES (29,2001);
INSERT INTO APPT_STAFF VALUES (30,4002);
INSERT INTO APPT_STAFF VALUES (31,2001);
INSERT INTO APPT_STAFF VALUES (32,3002);
INSERT INTO APPT_STAFF VALUES (52,9002);
INSERT INTO APPT_STAFF VALUES (53,2002);
INSERT INTO APPT_STAFF VALUES (54,5001);
```

INSERT INTO APPT_STAFF VALUES (55,10001);

INSERT INTO APPT_STAFF VALUES (56,10002);

INSERT INTO APPT_STAFF VALUES (57,10002);

INSERT INTO APPT_STAFF VALUES (58,1001);

INSERT INTO APPT_STAFF VALUES (59,4001);

-- FORMAT: INSERT INTO APPT_TREATMENT VALUES (APPTID,TREATNO);

INSERT INTO APPT_TREATMENT VALUES (1,202);

INSERT INTO APPT_TREATMENT VALUES (2,301);

INSERT INTO APPT_TREATMENT VALUES (3,101);

INSERT INTO APPT_TREATMENT VALUES (3,102);

INSERT INTO APPT_TREATMENT VALUES (5,801);

INSERT INTO APPT_TREATMENT VALUES (8,701);

INSERT INTO APPT_TREATMENT VALUES (9,102);

INSERT INTO APPT_TREATMENT VALUES (10,601);

INSERT INTO APPT_TREATMENT VALUES (11,701);

INSERT INTO APPT_TREATMENT VALUES (11,702);

INSERT INTO APPT_TREATMENT VALUES (13,802);

INSERT INTO APPT_TREATMENT VALUES (14,601);

INSERT INTO APPT_TREATMENT VALUES (15,201);

INSERT INTO APPT_TREATMENT VALUES (17,801);

INSERT INTO APPT_TREATMENT VALUES (18,301);

INSERT INTO APPT_TREATMENT VALUES (19,401);

INSERT INTO APPT_TREATMENT VALUES (20,101);

INSERT INTO APPT_TREATMENT VALUES (21,801);

INSERT INTO APPT_TREATMENT VALUES (23,902);

INSERT INTO APPT_TREATMENT VALUES (25,501);

INSERT INTO APPT_TREATMENT VALUES (26,1010);

INSERT INTO APPT_TREATMENT VALUES (29,201);

INSERT INTO APPT_TREATMENT VALUES (29,202);

INSERT INTO APPT_TREATMENT VALUES (30,401);

INSERT INTO APPT_TREATMENT VALUES (31,201);

INSERT INTO APPT_TREATMENT VALUES (54,501);

INSERT INTO APPT_TREATMENT VALUES (55,1010);

INSERT INTO APPT_TREATMENT VALUES (56,1010);

INSERT INTO APPT_TREATMENT VALUES (58,101);

INSERT INTO APPT_TREATMENT VALUES (58,102);

Task Three – Query The Dataset

Use MYSQL statements to query and modify the dataset as follows:

1. **Show details of all medical departments of hospital. (2 Marks)**

SELECT *

FROM DEPT;

```
5      -- Query 1:
6 •    SELECT *
7      FROM DEPT;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

DEPTNO	DNAME	LOC	BEDCAP	PHONE	EMAIL
1	Cardiology	First Floor	4	016466290	cardiology@hospital.ie
2	Dermatology	Second Floor	3	018093958	dermatology@hospital.ie
3	Endocrinology	Third Floor	2	014162661	endocrinology@hospital.ie
4	Gastroenterology	First Floor	2	018032366	gastroenterology@hospital.ie
5	Urology	Third Floor	4	012213125	urology@hospital.ie
6	Neurology	First Floor	3	014284482	neurology@hospital.ie
7	Oncology	Second Floor	2	016465639	oncology@hospital.ie
8	Orthopedics	First Floor	4	017972214	orthopedics@hospital.ie
9	Otolaryngology	Second Floor	2	018034859	otolaryngology@hospital.ie
10	Pediatrics	Fourth Floor	4	015106100	pediatrics@hospital.ie
11	Administration	Third Floor	NULL	012214000	admin@hospital.ie
NULL	NULL	NULL	NULL	NULL	NULL

2. **Find the contact details of patient "John Bob" whose patient ID is 5612.(2 Marks)**

SELECT *

FROM PATIENT_CONTACT

WHERE PATIENTID = 5612;

```
9       -- Query 2:
10 •    SELECT *
11      FROM PATIENT_CONTACT
12      WHERE PATIENTID = 5612;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

PATIENTID	PHONE	EMAIL	ADDRESS1	ADDRESS2	COUNTY	EIRCODE
5612	891112233	johnbob42@gmail.com	5 High Street	Coolmine Estate	Dublin	D15 F865
NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. **Update the patient ID number 3456 contact details. (2 Marks)**

```
UPDATE PATIENT_CONTACT
SET PHONE = 0893306504,
        EMAIL = 'saoirsemurph@gmail.com',
   ADDRESS1 = '3 Curragh Road',
   ADDRESS2 = 'Ratoath',
   COUNTY = 'Meath',
   EIRCODE = 'M78 H637'
WHERE PATIENTID = 3456;
SELECT *
FROM PATIENT_CONTACT
WHERE PATIENTID = 3456;
```

```
14      -- Query 3:
15 •    UPDATE PATIENT_CONTACT
16      SET PHONE = 0893306504,
17          EMAIL = 'saoirsemurph@gmail.com',
18          ADDRESS1 = '3 Curragh Road',
19          ADDRESS2 = 'Ratoath',
20          COUNTY = 'Meath',
21          EIRCODE = 'M78 H637'
22      WHERE PATIENTID = 3456;
23 •    SELECT *
24      FROM PATIENT_CONTACT
25      WHERE PATIENTID = 3456;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

PATIENTID	PHONE	EMAIL	ADDRESS1	ADDRESS2	COUNTY	EIRCODE
3456	893306504	saoirsemurph@gmail.com	3 Curragh Road	Ratoath	Meath	M78 H637
NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. **Schedule a new medical appointment. (2 Marks)**

```
INSERT INTO APPT (APPTID,DEPTNO,PATIENTID,DATE,TIME,STATUS)

VALUES(60,9,8457,'2025-11-20','10:00:00','Scheduled');


INSERT INTO APPT_STAFF (APPTID,STAFFID)

VALUES(60,9002);


INSERT INTO APPT_TREATMENT (APPTID,TREATNO)

VALUES(60,901);


SELECT *

FROM APPT A
```

JOIN APPT_STAFF AS ASF ON A.APPTID = ASF.APPTID

JOIN APPT_TREATMENT AS AT ON A.APPTID = AT.APPTID

WHERE A.APPTID = 60;

```
27      -- Query 4:
28      INSERT INTO APPT (APPTID,DEPTNO,PATIENTID,DATE,TIME,STATUS)
29      VALUES(60,9,8457,'2025-11-20','10:00:00','Scheduled');
30
31      INSERT INTO APPT_STAFF (APPTID,STAFFID)
32      VALUES(60,9002);
33
34      INSERT INTO APPT_TREATMENT (APPTID,TREATNO)
35      VALUES(60,901);
36
37      SELECT *
38      FROM APPT A
39      JOIN APPT_STAFF AS ASF ON A.APPTID = ASF.APPTID
40      JOIN APPT_TREATMENT AS AT ON A.APPTID = AT.APPTID
41      WHERE A.APPTID = 60;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

APPTID	DEPTNO	PATIENTID	DATE	TIME	STATUS	APPTID	STAFFID	APPTID	TREATNO
60	9	8457	2025-11-20	10:00:00	Scheduled	60	9002	60	901

5. **Find details of all patients who have Diabetes. (2 Marks)**

SELECT *

FROM MEDICAL_HISTORY

WHERE CONDITIONNAME = 'Diabetes';

SELECT p.*

FROM PATIENT p

JOIN MEDICAL_HISTORY mh ON p.PATIENTID = mh.PATIENTID

WHERE mh.CONDITIONNAME = 'Diabetes';

```
43      -- Query 5:
44 •    SELECT *
45      FROM MEDICAL_HISTORY
46      WHERE CONDITIONNAME = 'Diabetes';
47 •    SELECT p.*
48      FROM PATIENT p
49      JOIN MEDICAL_HISTORY mh ON p.PATIENTID = mh.PATIENTID
50      WHERE mh.CONDITIONNAME = 'Diabetes';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

PATIENTID	PFORENAME	PSURNAME	DOB	SEX
1312	Seamus	Kelly	1982-08-24	M
4765	Fiona	Murphy	1992-02-05	F
5235	Aidan	Smith	1974-12-08	M
5612	John	Bob	1942-07-31	M
8457	Patrick	Doherty	2019-05-09	M

6. **Display the first name and surname of all patients who are under the care of Dr. Kavin Smith. (2 Marks)**

```
SELECT P.PFORENAME, P.PSURNAME

FROM PATIENT P

JOIN APPT A ON P.PATIENTID = A.PATIENTID

JOIN APPT_STAFF AS ASF ON A.APPTID = ASF.APPTID

JOIN STAFF S ON ASF.STAFFID = S.STAFFID

WHERE S.SFORENAME = 'Kavin' AND S.SSURNAME = 'Smith';
```

```
52      -- Query 6:
53 •    SELECT P.PFORENAME, P.PSURNAME
54      FROM PATIENT P
55      JOIN APPT A ON P.PATIENTID = A.PATIENTID
56      JOIN APPT_STAFF AS ASF ON A.APPTID = ASF.APPTID
57      JOIN STAFF S ON ASF.STAFFID = S.STAFFID
58      WHERE S.SFORENAME = 'Kavin' AND S.SSURNAME = 'Smith';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

	PFORENAME	PSURNAME
▶	Ciaran	O Reilly
	Clodagh	Fitzgerald

7. **Change all patients in Orthopedics department to Orthopedic Surgery Department. (2 Marks)**

```
INSERT INTO DEPT (DEPTNO,DNAME,LOC,BEDCAP,PHONE,EMAIL)

VALUES(12,'Orthopedic Surgery','Fifth Floor', 4, '017312214','orthopedic.surgery@hospital.ie');


UPDATE DEPT_PATIENT

SET DEPTNO = 12

WHERE DEPTNO = 8;


SELECT *

FROM DEPT_PATIENT
```

WHERE DEPTNO = 12;

```
60      -- Query 7:
61      INSERT INTO DEPT (DEPTNO,DNAME,LOC,BEDCAP,PHONE,EMAIL)
62      VALUES(12,'Orthopedic Surgery','Fifth Floor', 4, '017312214','orthopedic.surgery@hospital.ie');
63
64      UPDATE DEPT_PATIENT
65      SET DEPTNO = 12
66      WHERE DEPTNO = 8;
67
68      SELECT *
69      FROM DEPT_PATIENT
70      WHERE DEPTNO = 12;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

DEPTNO	BEDNO	PATIENTID	ADMISSION_DATE
12	1	1312	2025-10-24
12	2	4325	2025-11-19
12	3	5312	2025-10-14
12	4	1432	2025-09-27
NULL	NULL	NULL	NULL

8. **Display the first name, surname, and hire date of all doctors sorted in descending order by hire date. (2 Marks)**

SELECT SFORENAME, SSURNAME, HIREDATE

FROM STAFF

WHERE JOB = 'Doctor'

ORDER BY HIREDATE DESC;

```
72      -- Query 8:
73      SELECT SFORENAME, SSURNAME, HIREDATE
74      FROM STAFF
75      WHERE JOB = 'Doctor'
76      ORDER BY HIREDATE DESC;
```

Result Grid | Filter Rows: | Export

SFORENAME	SSURNAME	HIREDATE
Alice	Nolan	2019-03-14
Fionn	McCartney	2016-09-23
Anna	Connolly	2013-08-02
Josh	Murphy	2012-10-16
Liam	Lennon	2011-12-07
Adam	Dunne	2011-08-28
Peter	Wang	2011-01-09
James	White	2010-05-25
Sarah	Brennan	2009-07-08
Kavin	Smith	2002-02-02

9. **Count how many patients are undergoing 'heart Surgery' treatment. (2 Marks)**

```
SELECT COUNT(DISTINCT APPT.PATIENTID) AS PatientCount
FROM APPT
JOIN APPT_TREATMENT ON APPT.APPTID = APPT_TREATMENT.APPTID
JOIN TREATMENT ON APPT_TREATMENT.TREATNO = TREATMENT.TREATNO
WHERE TREATMENT.TNAME = 'Heart surgery';
```

```
78      -- Query 9:
79      SELECT COUNT(DISTINCT APPT.PATIENTID) AS PatientCount
80      FROM APPT
81      JOIN APPT_TREATMENT ON APPT.APPTID = APPT_TREATMENT.APPTID
82      JOIN TREATMENT ON APPT_TREATMENT.TREATNO = TREATMENT.TREATNO
83      WHERE TREATMENT.TNAME = 'Heart surgery';
```



	PatientCount
▶	3

10. Add one new record to each table. (2 Marks)

INSERT INTO DEPT (DEPTNO, DNAME, LOC, BEDCAP, PHONE, EMAIL)

VALUES (13, 'Rehabilitation', 'Ground Floor', 2, '017234567', 'rehab@hospital.ie');

INSERT INTO TREATMENT (TREATNO, TNAME, DEPTNO)

VALUES (1301, 'Pelvic Floor Exercises', 13);

INSERT INTO STAFF (STAFFID, SFORENAME, SSURNAME, JOB, DEPTNO, PHONE, EMAIL, HIREDATE, DAYSOFF)

VALUES (13001, 'Emily', 'Johnson', 'Doctor', 13, 017233456, 'emily.johnson@hospital.ie', '2025-11-26', 2);

INSERT INTO PATIENT (PATIENTID, PFORENAME, PSURNAME, DOB, SEX)

VALUES (3749, 'Eoin', 'Taylor', '1985-07-22', 'M');

INSERT INTO DEPT_PATIENT (DEPTNO, BEDNO, PATIENTID, ADMISSION_DATE)

VALUES (13, 1, 10001, '2025-11-26');

INSERT INTO MEDICAL_HISTORY (PATIENTID, CONDITIONNAME, RECORDDATE, CONDITIONSTATUS)

VALUES (3749, 'Arthritis', '2025-11-26', 'Active');

INSERT INTO PATIENT_CONTACT (PATIENTID, PHONE, EMAIL, ADDRESS1, ADDRESS2, COUNTY, EIRCODE)

VALUES (3749, 017234567, 'eoin.taylor@gmail.com', '12 Ballentree Road', 'Treetown', 'Dublin', 'D01 Z872');

INSERT INTO APPT (APPTID, DEPTNO, PATIENTID, DATE, TIME, STATUS)

VALUES (61, 13, 3749, '2025-12-01', '10:00:00', 'Scheduled');

INSERT INTO APPT_STAFF (APPTID, STAFFID)

VALUES (61, 13001);

INSERT INTO APPT_TREATMENT (APPTID, TREATNO)

VALUES (61, 1301);

11. Delete one record from all tables. (2 Marks)

DELETE FROM DEPT

WHERE DEPTNO = 11;

DELETE FROM TREATMENT

WHERE TREATNO = 901;

DELETE FROM STAFF

WHERE STAFFID = 1111;

DELETE FROM PATIENT

WHERE PATIENTID = 1434;

DELETE FROM DEPT_PATIENT

WHERE PATIENTID = 1434;

DELETE FROM MEDICAL_HISTORY

WHERE PATIENTID = 1434;

DELETE FROM PATIENT_CONTACT

WHERE PATIENTID = 1434;

DELETE FROM APPT

WHERE APPTID = 9;

DELETE FROM APPT_STAFF

WHERE APPTID = 9;

DELETE FROM APPT_STAFF

WHERE APPT_TREATMENT = 9;

12. Find the total number of days off for all staff, order this by least days off. (2 Marks)

SELECT STAFFID, SFORENAME, SSURNAME, DAYSOFF

FROM STAFF

ORDER BY DAYSOFF ASC;

```
131     -- Query 12:
132 •   SELECT STAFFID, SFORENAME, SSURNAME, DAYSOFF
133     FROM STAFF
134     ORDER BY DAYSOFF ASC;
```

Result Grid | Filter Rows: | Edit:

STAFFID	SFORENAME	SSURNAME	DAYSOFF
2001	Adam	Dunne	1
7001	Kavin	Smith	1
1001	Sarah	Brennan	2
3001	Peter	Wang	2
13001	Emily	Johnson	2
3002	Sophia	Brown	3
3003	Benjamin	Taylor	3
8001	Fionn	McCartney	3
8003	Clara	Moore	3
9002	Jade	Hughes	3
10001	Josh	Murphy	3
2002	Olivia	Williams	4
4001	Anna	Connolly	4
6001	Alice	Nolan	4
6003	Ella	Davis	4
7004	Zoe	Scott	4
11001	Jason	Wade	4
1002	Emily	Johnson	5
6002	Ava	King	5
7005	Lucy	Martin	5
4002	Isabella	Maher	6
5001	Liam	Lennon	6
8002	Grace	Taylor	6
10003	Megan	Anderson	6
5002	Scarlett	Moore	7

13. Change the roll of all staff who have a job title Office worker to Administrator. (2 Marks)

```
UPDATE STAFF

SET JOB = 'Administrator'

WHERE JOB = 'Office worker';


SELECT *

FROM STAFF

WHERE JOB = 'Administrator';
```

```
136     -- Query 13:
137 •   UPDATE STAFF
138     SET JOB = 'Administrator'
139     WHERE JOB = 'Office worker';
140
141 •   SELECT *
142     FROM STAFF
143     WHERE JOB = 'Administrator';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

STAFFID	SFORENAME	SSURNAME	JOB	DEPTNO	PHONE	EMAIL	HIREDATE	DAYSOFF
11002	Holly	Jones	Administrator	11	13724327	holly.smith@hospital.ie	2023-03-30	8
11003	Nora	Dwyer	Administrator	11	17379744	nora.dwyer@hospital.ie	2024-02-02	11
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

14. Set the Orthopedics department phone number to be 01-7654321. (2 Marks)

```
UPDATE DEPT

SET NUMBER = '01-7654321'

WHERE DNAME = 'Orthopedics';
```

15. Set the chief executive officer (CEO) of the hospital to be called Michael Dean. (2 Marks)

```
UPDATE STAFF

SET SFORENAME = 'Michael',

        SSURNAME = 'Dean'

WHERE JOB = 'CEO';
```

16. Show all staff members who have been working there for longer than 3 years. (2 Marks)

SELECT STAFFID, SFORENAME, SSURNAME, JOB, HIREDATE

FROM STAFF

WHERE HIREDATE <= CURDATE() - INTERVAL 3 YEAR;

```
164     -- Query 16:
165 •   SELECT STAFFID, SFORENAME, SSURNAME, JOB, HIREDATE
166     FROM STAFF
167     WHERE HIREDATE <= CURDATE() - INTERVAL 3 YEAR;
```

Result Grid | Filter Rows: | Edit: | Expo

STAFFID	SFORENAME	SSURNAME	JOB	HIREDATE
1001	Sarah	Brennan	Doctor	2009-07-08
1002	Emily	Johnson	Nurse	2022-03-05
1111	Lisa	Costello	Cleaner	2018-06-18
2001	Adam	Dunne	Doctor	2011-08-28
3001	Peter	Wang	Doctor	2011-01-09
3002	Sophia	Brown	Nurse	2021-08-23
3003	Benjamin	Taylor	Healthcare Assistant	2021-06-14
4001	Anna	Connolly	Doctor	2013-08-02
5001	Liam	Lennon	Doctor	2011-12-07
5002	Scarlett	Moore	Nurse	2021-04-09
5003	Oliver	Johnson	Healthcare Assistant	2021-11-17
6001	Alice	Nolan	Doctor	2019-03-14
6002	Ava	King	Nurse	2022-06-14
7001	Kavin	Smith	Doctor	2002-02-02
7004	Zoe	Scott	Nurse	2021-09-01
7005	Lucy	Martin	Healthcare Assistant	2022-07-18
8001	Fionn	McCartney	Doctor	2016-09-23
8002	Grace	Taylor	Nurse	2022-11-22
8003	Clara	Moore	Healthcare Assistant	2021-05-11
9001	James	White	Doctor	2010-05-25
9002	Jade	Hughes	Nurse	2021-06-08
10001	Josh	Murphy	Doctor	2012-10-16
10002	Joseph	Clark	Nurse	2021-11-02
11001	Jason	Wade	CEO	2017-01-23
NULL	NULL	NULL	NULL	NULL

17. Show the detail of all patient names, medical history, appointment dates, times, and treatment (Hint: join operation). (2 Marks)

```
SELECT p.PFORENAME AS FirstName,
   p.PSURNAME AS LastName,
   mh.CONDITIONNAME AS MedicalCondition,
   mh.RECORDDATE AS RecordDate,
   a.DATE AS Appointment_Date,
   a.TIME AS Appointment_Time,
   t.TNAME AS Treatment_Name
FROM PATIENT p
LEFT JOIN MEDICAL_HISTORY mh ON p.PATIENTID = mh.PATIENTID
LEFT JOIN APPT a ON p.PATIENTID = a.PATIENTID
LEFT JOIN APPT_TREATMENT at ON a.APPTID = at.APPTID
LEFT JOIN TREATMENT t ON at.TREATNO = t.TREATNO
```

ORDER BY p.PFORENAME, p.PSURNAME, a.DATE;

```
169     -- Query 17:
170 •   SELECT p.PFORENAME AS FirstName,
171         p.PSURNAME AS LastName,
172         mh.CONDITIONNAME AS MedicalCondition,
173         mh.RECORDDATE AS RecordDate,
174         a.DATE AS Appointment_Date,
175         a.TIME AS Appointment_Time,
176         t.TNAME AS Treatment_Name
177     FROM PATIENT p
178     LEFT JOIN MEDICAL_HISTORY mh ON p.PATIENTID = mh.PATIENTID
179     LEFT JOIN APPT a ON p.PATIENTID = a.PATIENTID
180     LEFT JOIN APPT_TREATMENT at ON a.APPTID = at.APPTID
181     LEFT JOIN TREATMENT t ON at.TREATNO = t.TREATNO
182     ORDER BY p.PFORENAME, p.PSURNAME, a.DATE;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	MedicalCondition	RecordDate	Appointment_Date	Appointment_Time	Treatment_Name
Aidan	Smith	Obesity	2020-02-22	2025-10-18	09:00:00	Insulin
Aidan	Smith	Diabetes	2005-11-15	2025-10-18	09:00:00	Insulin
Aidan	Smith	Chronic Fatigue Syndrome	2019-07-20	2025-10-18	09:00:00	Insulin
Aoife	Byrne	Chronic Kidney Disease	2020-01-18	NULL	NULL	NULL
Brendan	Hughes	Urinary Tract Infection	2019-05-12	2025-10-31	15:00:00	NULL
Brendan	Hughes	Hypothyroidism	2015-10-05	2025-10-31	15:00:00	NULL
Bridget	Morris	Tonsillitis	2021-01-05	2025-11-21	12:30:00	Tonsillectomy
Bridget	Morris	Tonsillitis	2021-01-05	2025-11-22	10:30:00	NULL
Caitriona	Murphy	Chronic Back Pain	2017-09-13	2025-11-19	09:30:00	Cartilage repair
Ciaran	O Reilly	Prostate Cancer	2019-09-15	2025-11-14	14:30:00	Chemotherapy
Clodagh	Fitzgerald	Cervical Cancer	2018-06-17	2025-11-06	12:00:00	Chemotherapy
Clodagh	Fitzgerald	Cervical Cancer	2018-06-17	2025-11-06	12:00:00	Radiotherapy
Conor	Murray	Arrhythmia	2019-05-09	2025-11-12	11:00:00	Heart surgery

18. Identify how many patients have the word road in their address. (1 Marks)

SELECT COUNT(*)

FROM PATIENT_CONTACT

WHERE (ADDRESS1 LIKE '%road%' OR ADDRESS2 LIKE '%road%');

```
184     -- Query 18:
185 •   SELECT COUNT(*)
186     FROM PATIENT_CONTACT
187     WHERE (ADDRESS1 LIKE '%road%' OR ADDRESS2 LIKE '%road%');
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

COUNT(*)
6

19. Create a view that will show the result of a query drawing information from three tables at once. (3 marks)

```
CREATE VIEW Patient_Dept_Contact AS
SELECT p.PFORENAME AS FirstName,
   p.PSURNAME AS LastName,
   dp.BEDNO AS BedNumber,
   dp.ADMISSION_DATE AS Admission_Date,
   pc.PHONE AS Phone,
   pc.EMAIL AS Email,
   pc.ADDRESS1 AS Address1,
   pc.ADDRESS2 AS Address2,
   pc.COUNTY AS County,
   pc.EIRCODE AS Eircode
FROM PATIENT p
JOIN DEPT_PATIENT dp ON p.PATIENTID = dp.PATIENTID
JOIN PATIENT_CONTACT pc ON p.PATIENTID = pc.PATIENTID;
```

20. Drop all information contained in the appointments relation. (2 Marks)

```
DROP TABLE APPT;
```